MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 4705: INFORMATION SYSTEMS AUDIT

END OF SEMESTER EXAMINATION

DATE: MARCH, 2020 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

1. Answer **ALL** questions.
2. Marks for each question are indicated.
3. Write your answers clearly.
4. Indicate the question number for each answer.
5. This paper has a total of **EIGHT** Questions
6. This paper contains **THREE** pages, without including the cover page. Please check

1. Suppose your ICT organization requires some Auditing.
    a. Outline any **three** reasons why an organization may require services of an auditor. **[3 marks]**
    b. Describe any **three** characteristics of a good IS Auditor. **[6 marks]**
    c. Give any **three** job descriptions that you may include in an advert to hire such an auditor. **[3 marks]**
    d. Explain the auditing process. **[7 marks]**
    e. Describe any **three** IS auditing standards and guidelines. **[6 marks]**

2. Most of the auditing activities happen within the Information System (IS) Audit function in the organization where the IS Auditor has his duties, rights, limitations, as well as professional code of ethics to consider.
    a. Describe the role of the IS Audit function. **[4 marks]**
    b. Explain the mandate of the Information Systems Auditor in performing the IS Audit function. **4 marks]**
    c. Discuss how you can implement each of the following kinds of auditing:
        i. Auditing of information security management frameworks. **[4 marks]**
        ii. Auditing network infrastructure security. **[4 marks]**
        iii. Auditing systems operations. **[4 marks]**

3. Describe:
    a. Any **two** rights of an IS Auditor. **[4 marks]**
    b. Any **two** limitations of an IS Auditor. **[4 marks]**
    c. Any **two** code of ethics to be observed by the IS Auditor. **[4 marks]**
    d. Discuss how different the rights, limitations, and code of ethics of an Internal IS Auditor are from those of an External IS Auditor. **[8 marks]**

4. Corporate governance and IT governance are important elements to be considered in IS Auditing.
    a. Describe the role that corporate governance plays in IS Auditing. **[3 marks]**
    b. Explain any **two** duties of board of directors IS Auditing. **[3 marks]**

c. Apart from independence of the Auditor, explain how you can achieve the independence of the different levels of an IS organization hierarchy. **[4 marks]**

d. Explain how any **two** IS organization structures affect IS Auditing. **[6 marks]**

e. Explain how an IS Auditor can undertake the auditing of IT governance structure. **[4 marks]**

5. Every information system is associated with some risks of different degrees.

a. Describe Risk management. **[3 marks]**

b. Using the CIA Triad as your reference, discuss any **three** information systems risks and ways of mitigating each of the risks. **[9 marks]**

c. Describe the components of an organizational framework for security and control. **[7 marks]**

d. Describe a maturity model. **[3 marks]**

e. Explain the goal of a process maturity framework. **[3 marks]**

6. ITIL is all about Service, Service, Service.

a. Discuss what the above statement means in line with IT Service Delivery, Infrastructure and Operations Support. **[4 marks]**

b. Differentiate the duties of an IT service desk from those of an IT help desk. **[4 marks]**

c. Discuss the IT Service Life Cycle? **[8 marks]**

7. Suppose you are requested to provide IS security solutions to RTZ insurance company following the principle that IS security measures are based on value of each asset.

a. Describe any **three** Information Systems asset type and their value. **[6 marks]**

b. Describe any **three** Best practices for logon IDs and passwords. **[6 marks]**

c. Differentiate General operating systems access control functions from application-level access control functions. **[6 marks]**

d. Describe how each of the following elements contribute in safeguarding the security of the information assets

i. Dimilitarized zone. **[3 marks]**

ii. Server farms. **[3 marks]**

iii. Load balancer. **[3 marks]**

8. Suppose you have been employed as an Information Systems Auditor for Zanga zatha Bank where you have been included in a team responsible for initiating a business continuity, and disaster recovery plan. In your opening meeting, your Chief Information Officer concludes with the statement that goes: "for the survival of any organisation, a disaster recovery plan is a ***must*** have".

   a. Discuss the steps of coming up with a disaster recovery plan. **[6 marks]**

   b. In implementing the systems or capabilities required to deliver the plan, there are three primary types of sites to deal with situations where the data center or main facility are unavailable. Describe these **three** sites. **[6 marks]**

   c. Is business continuity the same as disaster recovery planning? Discuss your answer. **[4 marks]**

   d. Explain how you can audit business continuity. **[4 marks]**

End of Question Paper